

```
QEMU (Win-2)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 180.0.0.10
Pinging 180.0.0.10 with 32 bytes of data:
Reply from 180.0.0.10: bytes=32 time=7ms TTL=60
Reply from 180.0.0.10: bytes=32 time=3ms TTL=60
Ping statistics for 180.0.0.10:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 7ms, Average = 5ms
Control-C
^C
C:\Documents and Settings\user>ping 180.0.0.20
Pinging 180.0.0.20 with 32 bytes of data:
Reply from 180.0.0.20: bytes=32 time=5ms TTL=60
Reply from 180.0.0.20: bytes=32 time=3ms TTL=60
Ping statistics for 180.0.0.20:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 5ms, Average = 4ms
Control-C
^C
C:\Documents and Settings\user>ping 192.52.234.100
Pinging 192.52.234.100 with 32 bytes of data:
Reply from 192.52.234.100: bytes=32 time=6ms TTL=60
Reply from 192.52.234.100: bytes=32 time=5ms TTL=60
Ping statistics for 192.52.234.100:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 6ms, Average = 5ms
Control-C
^C
C:\Documents and Settings\user>ping 192.52.234.200
Pinging 192.52.234.200 with 32 bytes of data:
Reply from 192.52.234.200: bytes=32 time=4ms TTL=60
Reply from 192.52.234.200: bytes=32 time=4ms TTL=60
Ping statistics for 192.52.234.200:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 4ms, Average = 4ms
Control-C
^C
C:\Documents and Settings\user>
C:\Documents and Settings\user>
```

```
QEMU (Win-1)
C:\WINDOWS\system32\cmd.exe
My
Pinging 180.0.0.10 with 32 bytes of data:
Reply from 180.0.0.10: bytes=32 time=6ms TTL=60
Reply from 180.0.0.10: bytes=32 time=4ms TTL=60
My
Ping statistics for 180.0.0.10:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 6ms, Average = 5ms
Control-C
^C
C:\Documents and Settings\user>ping 180.0.0.20
My
Pinging 180.0.0.20 with 32 bytes of data:
Reply from 180.0.0.20: bytes=32 time=2ms TTL=60
Reply from 180.0.0.20: bytes=32 time=4ms TTL=60
Rs
Ping statistics for 180.0.0.20:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 3ms
Control-C
^C
C:\Documents and Settings\user>ping 192.52.234.100
I
Pinging 192.52.234.100 with 32 bytes of data:
E
Reply from 192.52.234.100: bytes=32 time=3ms TTL=60
Reply from 192.52.234.100: bytes=32 time=3ms TTL=60
Ping statistics for 192.52.234.100:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
A
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 3ms, Average = 3ms
Control-C
^C
C:\Documents and Settings\user>ping 192.52.234.200
Old
Pinging 192.52.234.200 with 32 bytes of data:
Reply from 192.52.234.200: bytes=32 time=6ms TTL=60
Reply from 192.52.234.200: bytes=32 time=3ms TTL=60
Ping statistics for 192.52.234.200:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 6ms, Average = 4ms
Control-C
^C
C:\Documents and Settings\user>_
```

QEMU (Win-3)



```
My C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 180.0.0.20
Pinging 180.0.0.20 with 32 bytes of data:
Reply from 180.0.0.20: bytes=32 time=4ms TTL=60
My Reply from 180.0.0.20: bytes=32 time=5ms TTL=60
Reply from 180.0.0.20: bytes=32 time=4ms TTL=60
Ping statistics for 180.0.0.20:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 5ms, Average = 4ms
My Control-C
^C
C:\Documents and Settings\user>ping 180.0.0.10
Re Pinging 180.0.0.10 with 32 bytes of data:
Reply from 180.0.0.10: bytes=32 time=3ms TTL=60
Reply from 180.0.0.10: bytes=32 time=4ms TTL=60
Ping statistics for 180.0.0.10:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms
I Control-C
B ^C
C:\Documents and Settings\user>ping 192.52.234.100
A Pinging 192.52.234.100 with 32 bytes of data:
Reply from 192.52.234.100: bytes=32 time=3ms TTL=60
Reply from 192.52.234.100: bytes=32 time=4ms TTL=60
Ping statistics for 192.52.234.100:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms
Old Control-C
^C
C:\Documents and Settings\user>ping 192.52.234.200
Pinging 192.52.234.200 with 32 bytes of data:
Reply from 192.52.234.200: bytes=32 time=3ms TTL=60
Reply from 192.52.234.200: bytes=32 time=4ms TTL=60
Ping statistics for 192.52.234.200:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms
Control-C
^C
C:\Documents and Settings\user>_
```

```
QEMU (Win-4)
Ctrl+Alt+Del to host
C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ping 180.0.0.10

Pinging 180.0.0.10 with 32 bytes of data:

Reply from 180.0.0.10: bytes=32 time=5ms TTL=60
Reply from 180.0.0.10: bytes=32 time=4ms TTL=60
Reply from 180.0.0.10: bytes=32 time=4ms TTL=60

Ping statistics for 180.0.0.10:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 5ms, Average = 4ms
Control-C
^C
C:\Documents and Settings\user>ping 180.0.0.20

Pinging 180.0.0.20 with 32 bytes of data:

Reply from 180.0.0.20: bytes=32 time=4ms TTL=60
Reply from 180.0.0.20: bytes=32 time=4ms TTL=60

Ping statistics for 180.0.0.20:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 4ms, Average = 4ms
Control-C
^C
C:\Documents and Settings\user>ping 192.52.234.100

Pinging 192.52.234.100 with 32 bytes of data:

Reply from 192.52.234.100: bytes=32 time=3ms TTL=60
Reply from 192.52.234.100: bytes=32 time=3ms TTL=60

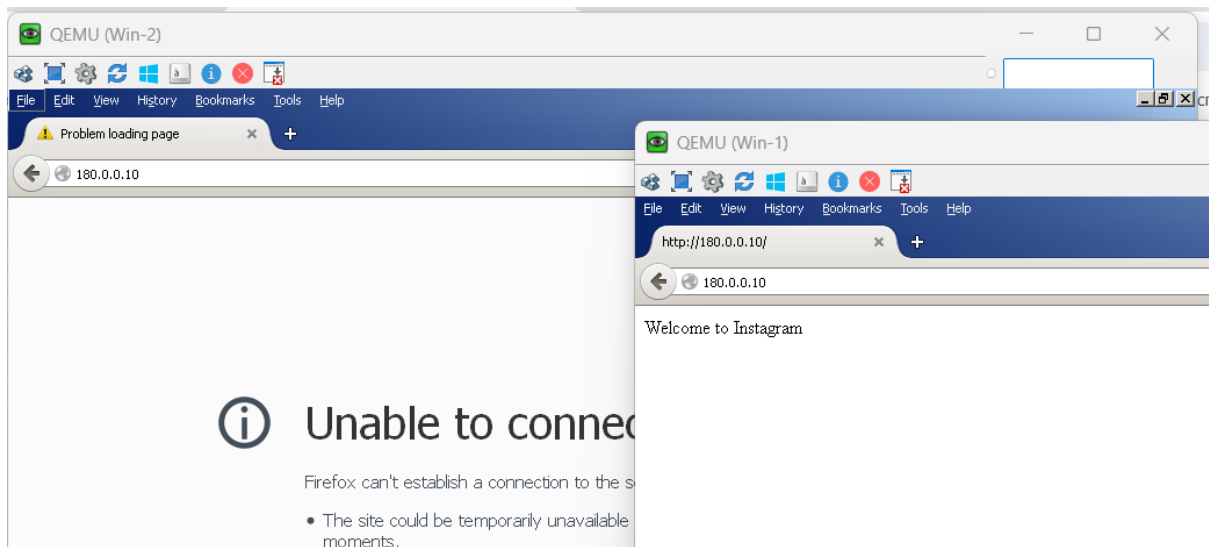
Ping statistics for 192.52.234.100:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 3ms, Average = 3ms
Control-C
^C
C:\Documents and Settings\user>ping 192.52.234.200

Pinging 192.52.234.200 with 32 bytes of data:

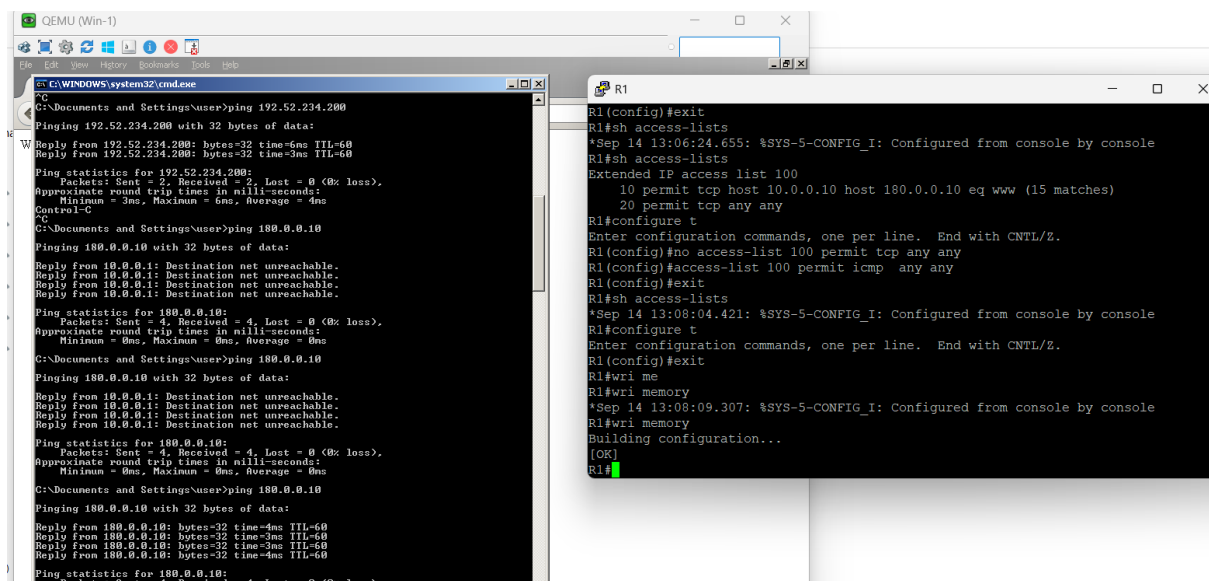
Reply from 192.52.234.200: bytes=32 time=4ms TTL=60
Reply from 192.52.234.200: bytes=32 time=3ms TTL=60

Ping statistics for 192.52.234.200:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms
Control-C
^C
C:\Documents and Settings\user>_
```

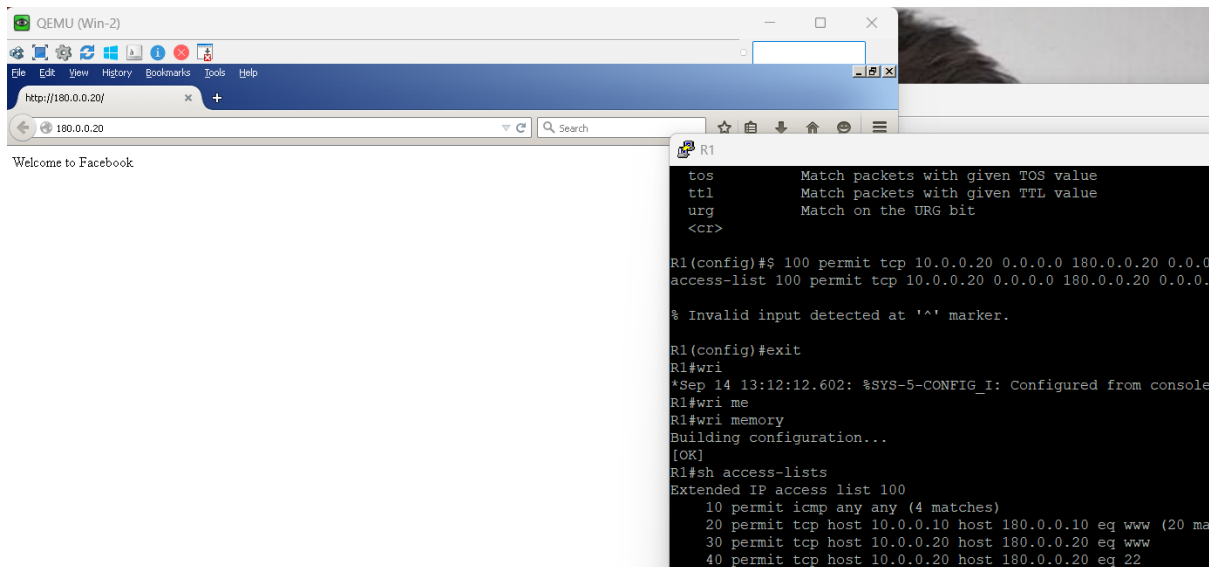
Scenario : 1 == Allow 10.0.0.0 to communicate only instagram server using port 80



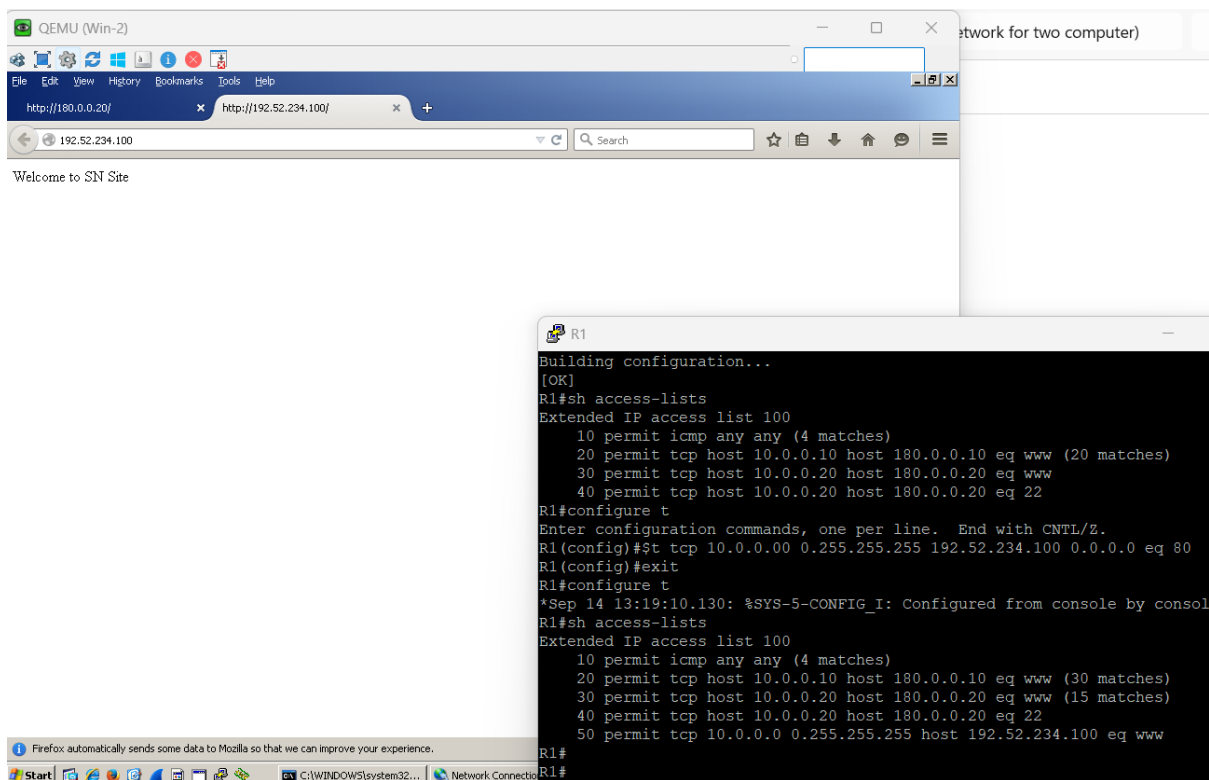
Scenario : 2 == Allow ping communication from 10.0.0.10 to instagram server



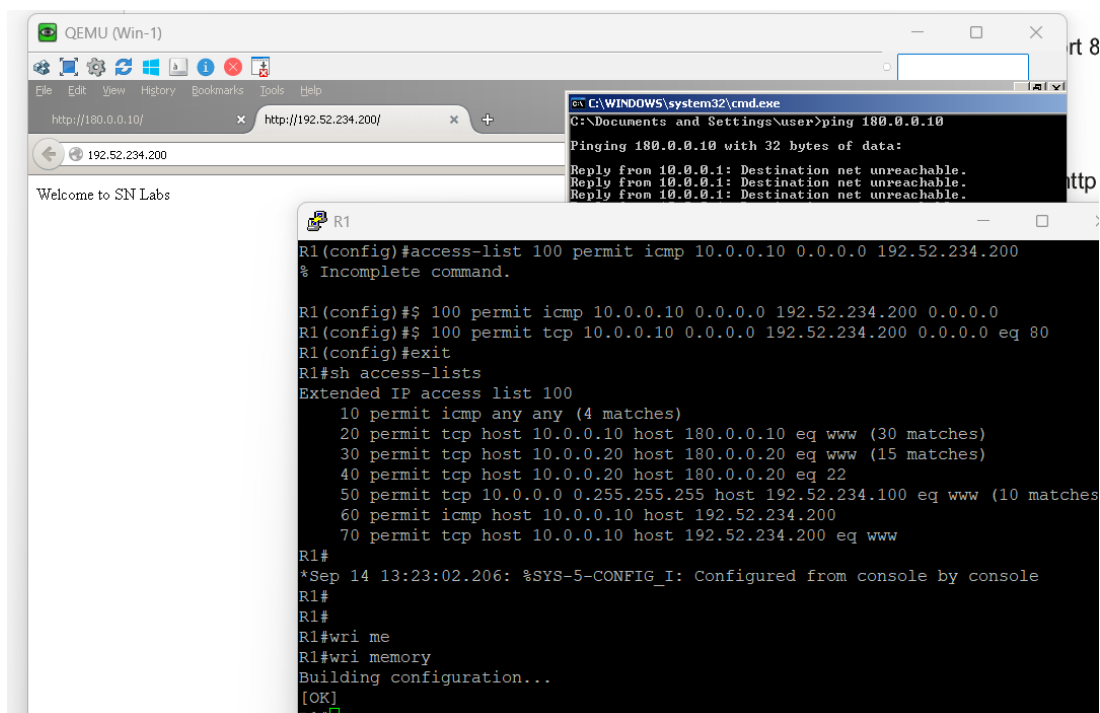
Scenario : 3 == Allow 10.0.0.20 to communicate facebook server using ssh & http



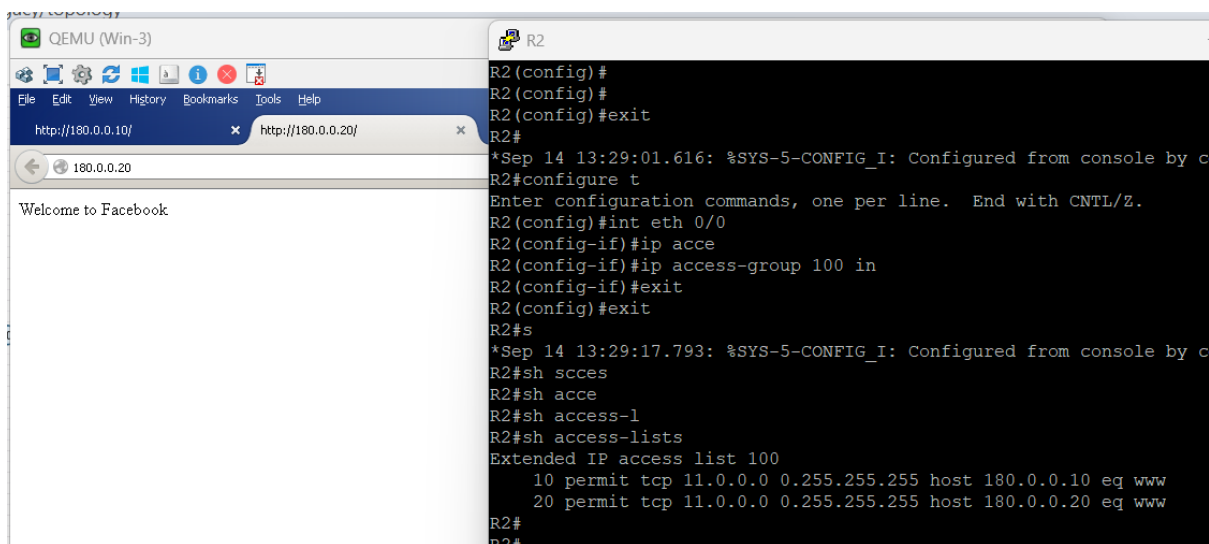
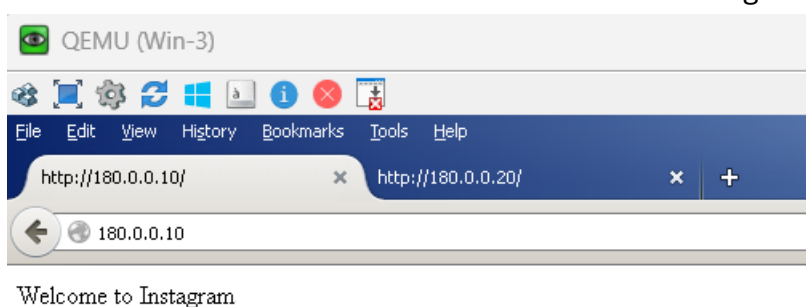
Scenario : 4 === Allow 10.0.0.0 to communicate sns site server using http



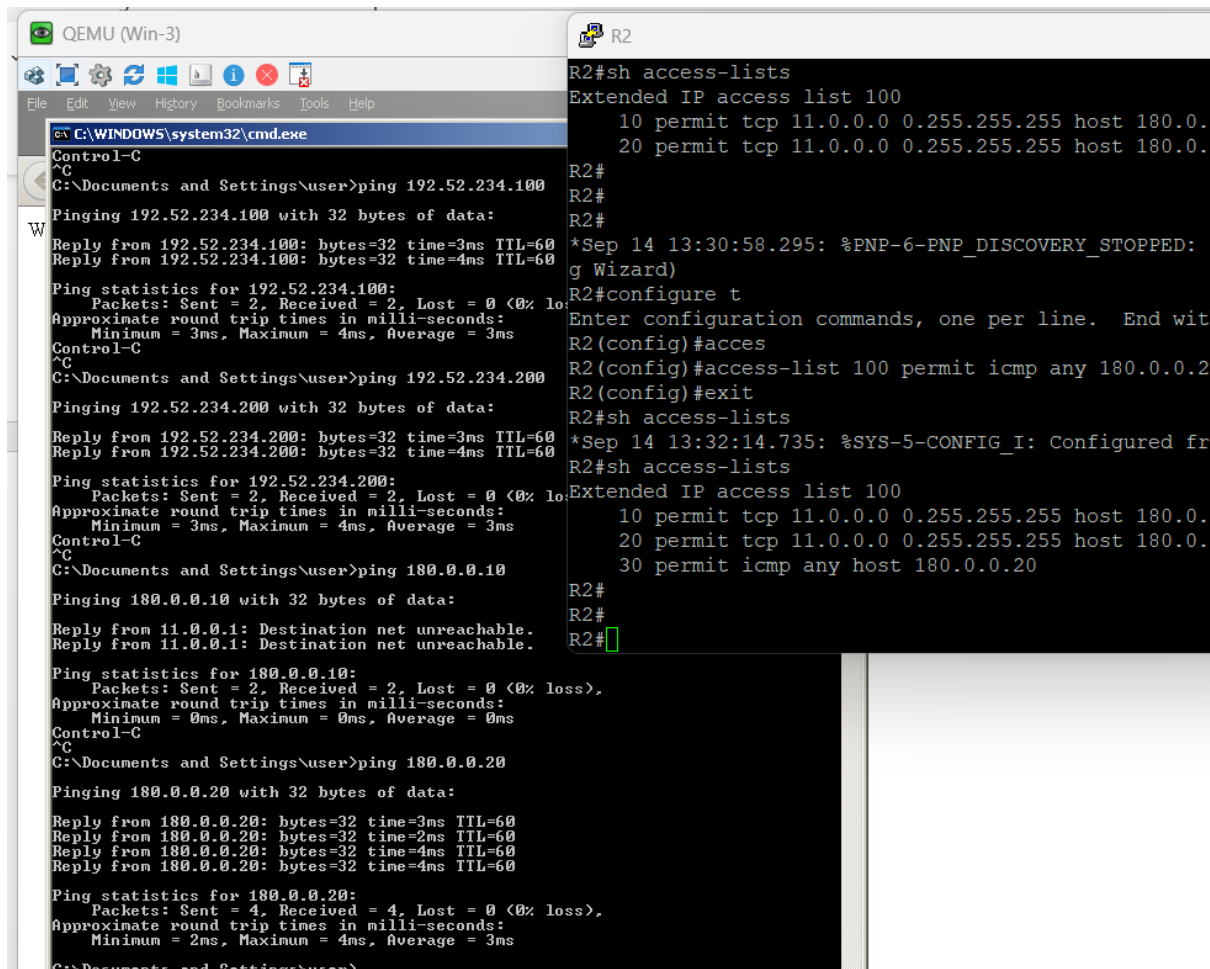
Scenario : 5 === Allow 10.0.0.10 to communicate snlabs server using http & icmp



Scenario : 6 === Allow 11.0.0.0 to communicate instagram & facebook server using http



Scenario : 7 === Allow 11.0.0.10 to ping facebook server

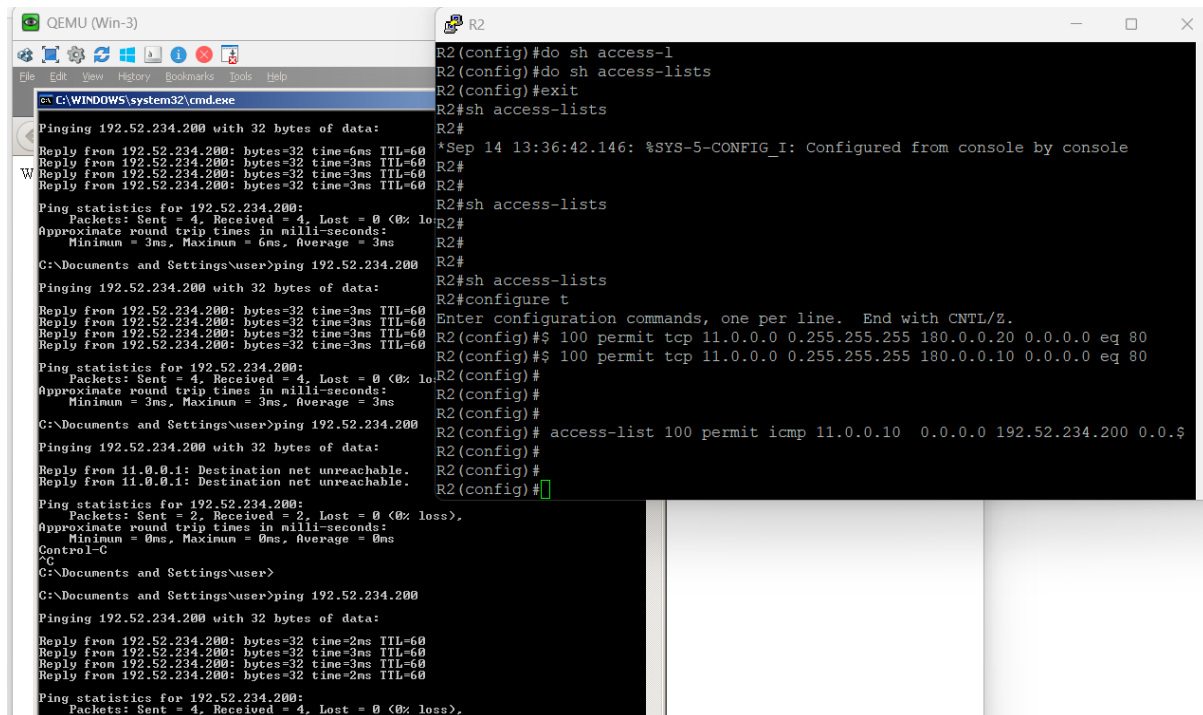


```
QEMU (Win-3)
File Edit View History Bookmarks Tools Help
C:\WINDOWS\system32\cmd.exe
Control-C
^C
C:\Documents and Settings\user>ping 192.52.234.100
Pinging 192.52.234.100 with 32 bytes of data:
Reply from 192.52.234.100: bytes=32 time=3ms TTL=60
Reply from 192.52.234.100: bytes=32 time=4ms TTL=60
Ping statistics for 192.52.234.100:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms
Control-C
^C
C:\Documents and Settings\user>ping 192.52.234.200
Pinging 192.52.234.200 with 32 bytes of data:
Reply from 192.52.234.200: bytes=32 time=3ms TTL=60
Reply from 192.52.234.200: bytes=32 time=4ms TTL=60
Ping statistics for 192.52.234.200:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms
Control-C
^C
C:\Documents and Settings\user>ping 180.0.0.10
Pinging 180.0.0.10 with 32 bytes of data:
Reply from 11.0.0.1: Destination net unreachable.
Reply from 11.0.0.1: Destination net unreachable.
Ping statistics for 180.0.0.10:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
Control-C
^C
C:\Documents and Settings\user>ping 180.0.0.20
Pinging 180.0.0.20 with 32 bytes of data:
Reply from 180.0.0.20: bytes=32 time=3ms TTL=60
Reply from 180.0.0.20: bytes=32 time=2ms TTL=60
Reply from 180.0.0.20: bytes=32 time=4ms TTL=60
Reply from 180.0.0.20: bytes=32 time=4ms TTL=60
Ping statistics for 180.0.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 3ms
C:\Documents and Settings\user>

R2
R2#sh access-lists
Extended IP access list 100
 10 permit tcp 11.0.0.0 0.255.255.255 host 180.0.
 20 permit tcp 11.0.0.0 0.255.255.255 host 180.0.
R2#
R2#
R2#
*Sep 14 13:30:58.295: %PNP-6-PNP_DISCOVERY_STOPPED:
g Wizard)
R2#configure t
Enter configuration commands, one per line. End with
R2(config)#access
R2(config)#access-list 100 permit icmp any 180.0.0.2
R2(config)#exit
R2#sh access-lists
*Sep 14 13:32:14.735: %SYS-5-CONFIG_I: Configured fr
R2#sh access-lists
Extended IP access list 100
 10 permit tcp 11.0.0.0 0.255.255.255 host 180.0.
 20 permit tcp 11.0.0.0 0.255.255.255 host 180.0.
 30 permit icmp any host 180.0.0.20
R2#
R2#
R2#
```

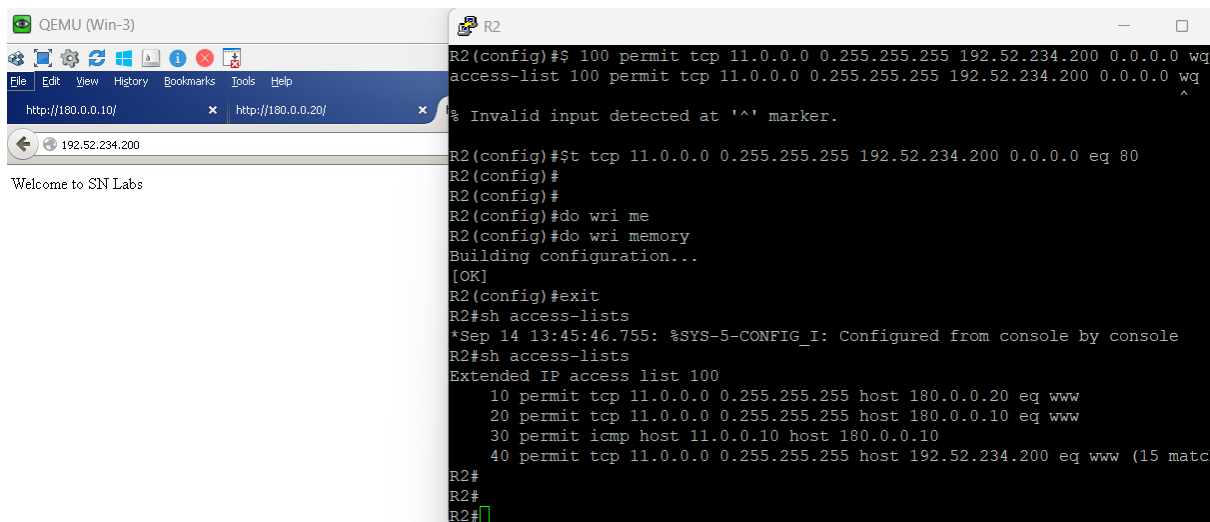


Scenario : 8 === Allow 11.0.0.10 to ping snlab server



The screenshot shows a QEMU (Win-3) window with two panes. The left pane displays a Windows command prompt running the command `C:\WINDOWS\system32\cmd.exe`. It shows a successful ping to 192.52.234.200 with 32 bytes of data, 4 packets sent, and 4 received. The right pane shows a Linux terminal (R2) running the following commands: `R2(config)#do sh access-l`, `R2(config)#do sh access-lists`, `R2(config)#exit`, `R2#sh access-lists`, `R2#`, `R2#sh access-lists`, `R2#`, `R2#sh access-lists`, `R2#configure t`, `R2(config)#`, `R2(config)#$ 100 permit tcp 11.0.0.0 0.255.255.255 180.0.0.20 0.0.0.0 eq 80`, `R2(config)#$ 100 permit tcp 11.0.0.0 0.255.255.255 180.0.0.10 0.0.0.0 eq 80`, `R2(config)#`, `R2(config)#`, `R2(config)# access-list 100 permit icmp 11.0.0.10 0.0.0.0 192.52.234.200 0.0.0.0`, `R2(config)#`, `R2(config)#`, and `R2(config)#`. The terminal also shows a message: `*Sep 14 13:36:42.146: %SYS-5-CONFIG_I: Configured from console by console`.

Scenario : 9 === Allow 11.0.0.0 to access snlab server



The screenshot shows a QEMU (Win-3) window with two panes. The left pane displays a web browser with the address bar showing `http://180.0.0.10/` and `http://180.0.0.20/`. The right pane shows a Linux terminal (R2) running the following commands: `R2(config)#$ 100 permit tcp 11.0.0.0 0.255.255.255 192.52.234.200 0.0.0.0 wq`, `access-list 100 permit tcp 11.0.0.0 0.255.255.255 192.52.234.200 0.0.0.0 wq`, `% Invalid input detected at '^' marker.`, `R2(config)#$t tcp 11.0.0.0 0.255.255.255 192.52.234.200 0.0.0.0 eq 80`, `R2(config)#`, `R2(config)#`, `R2(config)#do wri me`, `R2(config)#do wri memory`, `Building configuration...`, `[OK]`, `R2(config)#exit`, `R2#sh access-lists`, `*Sep 14 13:45:46.755: %SYS-5-CONFIG_I: Configured from console by console`, `R2#sh access-lists`, `Extended IP access list 100`, `10 permit tcp 11.0.0.0 0.255.255.255 host 180.0.0.20 eq www`, `20 permit tcp 11.0.0.0 0.255.255.255 host 180.0.0.10 eq www`, `30 permit icmp host 11.0.0.10 host 180.0.0.10`, `40 permit tcp 11.0.0.0 0.255.255.255 host 192.52.234.200 eq www (15 match`, `R2#`, `R2#`, and `R2#`.

Scenario : 10 === Deny 11.0.0.0 to access 10.0.0.0

By default its was deny due to implicit rule